

Benno Käch | Curriculum Vitae

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Professional Summary

Expert in machine learning, data analysis, and statistics, with a focus on generative modeling for discrete data and point clouds. Proven ability to deliver impactful results across industries, from optimizing pricing models to deploying ML solutions in manufacturing and research. Eager to leverage deep technical expertise to solve complex challenges and drive innovation in dynamic, results-oriented environments.

Experience

Simpego **Zurich, Switzerland**
Data Analyst *2017–2020*

- Gained initial experience in machine learning by helping to set up, optimize, and deploy a boosted decision tree-based pricing model in R.
- Developed monitoring tools to detect market changes, based on quantile-based sampling compatible with both categorical and numerical data.
- Replaced outsourced data mining with an in-house web scraper using Selenium and BeautifulSoup.

ETH Zurich, IWF **Zurich, Switzerland**
Research Assistant *2020–2021*

- Contributed to the implementation of ML-based process automation in manufacturing.
- Supervised two master's theses on head pose detection and emotion recognition.
- Communicated academic findings to industry partners and implemented them on production machinery.

Deutsches Elektronen-Synchrotron (DESY) **Hamburg, Germany**
Researcher (PhD/Postdoc) *2021–2024*

- Developed generative models for applications in high-energy physics.
- Worked with GANs, flow matching, and normalizing flows for point cloud generation, achieving results competitive with state of the art.
- Built signal-background classifiers for highly imbalanced datasets.
- Analyzed petabytes of data recorded by the CMS detector, conducting advanced statistical tests to compare observations with theoretical expectations.

In Virtuo Laboratories **Lugano, Switzerland**
Head of Machine Learning *2025–Present*

- Within one year of transitioning from high-energy physics, developed a discrete flow matching model for fragmented SMILES trained on billions of compounds. Proposed a novel reinforcement learning-based algorithm for downstream oracle optimization benchmarks. Publication accepted at the NeurIPS Drug Discovery Workshop and submitted to ICLR 2026 main track.
- Surpassed NVIDIA's GenMol across all benchmarks (de novo generation, fragmented constrained generation, PMO benchmark, lead optimization).
- Led additional projects including active learning and surrogate docking on large molecular databases.

Education

University of Hamburg

Ph.D. in Particle Physics

Magna Cum Laude

2021–2024

ETH Zurich

M.Sc. in Physics

2019–2020

ETH Zurich

B.Sc. in Physics

2015–2018

Selected Scientific Record

NeurIPS 2022, ML4PS: *Point Cloud Generation using Transformer Encoders and Normalizing Flows*, co-authored with Dirk Krücker and Isabell Melzer Pellmann

NeurIPS 2023, ML4PS: *Pay Attention to Mean-Fields for Point Cloud Generation*, co-authored with Dirk Krücker and Isabell Melzer Pellmann

PhD Thesis: *Generative Modelling in High-Energy Physics*

Neurips 2025, AI4D3: Refine Drugs, Don't Complete Them: Uniform-Source Discrete Flows for Fragment-Based Drug Discovery

Technical Skills

Programming Languages: Python (expert), R, C++, MATLAB, Scala, SQL, Bash scripting, CSS, HTML, Markdown, $\LaTeX$

Machine Learning : PyTorch, Lightning, scikit-learn, H2O, CatBoost, Weights & Biases, Ray

Cloud Computing: AWS Glue, S3

Data Analysis: pandas

Data Visualization: matplotlib, seaborn, scikit-learn

Web Scraping: Selenium, BeautifulSoup

Resource Management: SLURM

Data Structure and Algorithms: LeetCode

Personal Information

Nationality: Switzerland

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Languages

German: Native

English: Fluent

French: Conversational

Italian: Basic